

NENA: BIDIRECTIONAL LIVE-CALLING VOIP SYSTEM ARCHITECTURE

A High-Impact Technical Framework for Real-Time Swahili Sign-to-Speech and Speech-to-Sign Communication Relay

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1. INTRODUCTION & THEORETICAL FOUNDATIONS

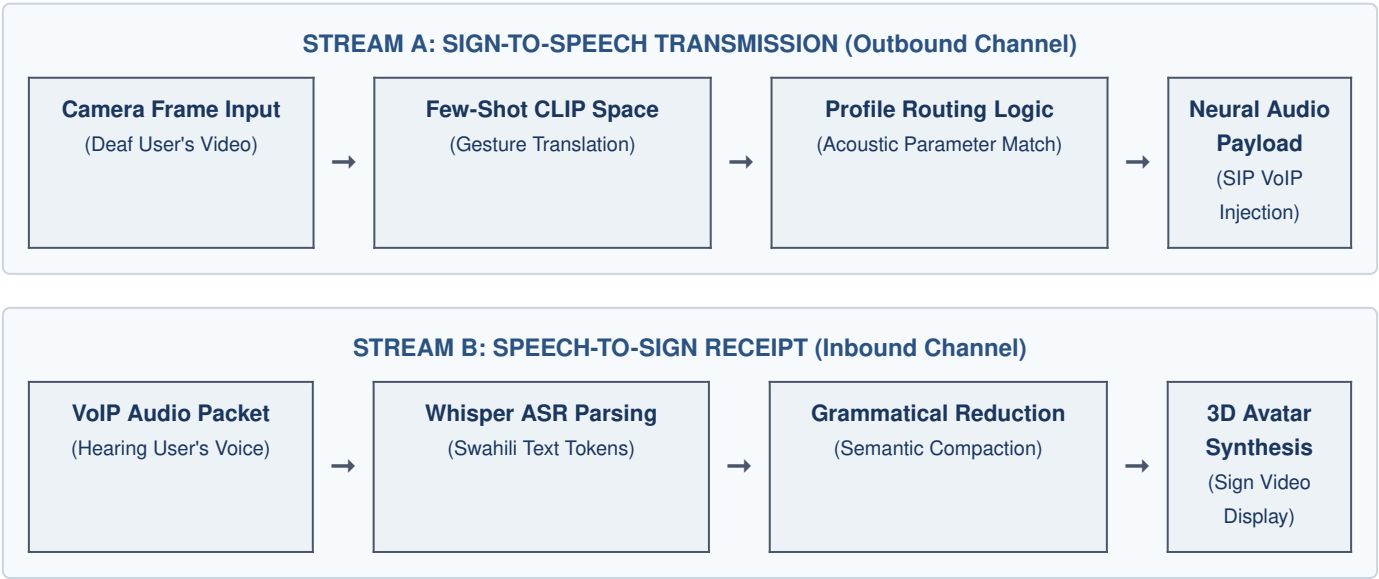
Deaf and hard-of-hearing individuals in East Africa face profound communication barriers, particularly during telephonic interactions. Standard mobile communication networks (GSM) rely exclusively on real-time acoustic channels, completely rendering remote interaction impossible for sign-language users without human visual interpreters.

This technical document details the design specifications for **NENA**, an AI-powered bidirectional mobile communication ecosystem. NENA integrates automated **Sign-to-Speech** and **Speech-to-Sign** computational engines into an active Voice-over-IP (VoIP) stream.

The visual translation layer avoids exhaustive end-to-end dataset generation by building upon *Contrastive Vision-Language Alignment Frameworks* and *Few-Shot Learning Paradigms*. By using deep encoders pre-trained on text-image pairs, the architecture maps dynamic sign language video tokens directly to a shared multimodal semantic embedding space. This allows the model to map unfamiliar Swahili gestures with exceptional precision from only a handful of calibration examples.

2. COMPLETE BIDIRECTIONAL SYSTEM PIPELINE

The platform operates on a split-stream asynchronous model, breaking down communication boundaries during live audio-video calls. The pipeline handles two separate processing paths concurrently:



3. MATHEMATICAL FORMALIZATION OF CONTRASTIVE FEW-SHOT GESTURE RECOGNITION

The spatial-temporal boundaries of dynamic Swahili gestures are captured through continuous mobile camera streams. The frame sequencing block takes a video clip V and extracts feature sets through a Vision Transformer network encoder f_v . Text phrases are projected concurrently via a text encoder f_t .

Let $z_v = f_v(V)$ be the target visual vector and $z_t = f_t(T)$ be the corresponding text vector. To execute low-sample categorization, NENA utilizes Prototypical Distances. Given a specific query gesture vector q , the class probability is calculated using the distance to class center-points c_k across a shared metric surface:

$$P(y = k | q) = \exp(-d(q, c_k)) / \sum_j \exp(-d(q, c_j))$$

Where $d(\cdot)$ corresponds to the Euclidean distance calculation. This empowers the core application layer to perform sub-second sign matching without exhaustive client-side model updates.

4. ADVANCED CORE COMPONENTS

4.1. The Swahili Pronominal Morphological Resolution (User Profile Settings)

In Swahili grammar, subject prefixes lack gender distinctions. For instance, the expression "Yeye *anasoma*" equates to both "He is reading" and "She is reading." Because the text representations lack semantic gender cues, downstream automated Text-to-Speech (TTS) pipelines face fundamental voice-assignment errors.

NENA bypasses this issue entirely by decoupling biological tracking from the system. Instead, the framework maps tokens directly through an **Application Settings Profile State**. The user chooses their preferred acoustic identity during initial account creation:

- **Masculine Parameter Engine:** Sets output voice targets to a deeper, low-frequency acoustic profile (e.g., sw-TZ-Wavenet-B).
- **Feminine Parameter Engine:** Sets output voice targets to a higher-frequency, melodic acoustic profile (e.g., sw-TZ-Wavenet-A).

4.2. Live Call VoIP Engine & Session Initiation Layer

The live audio calling workflow utilizes Web Real-Time Communication (WebRTC) pipelines joined to a Session Initiation Protocol (SIP) gateway. This configuration permits smooth calls between standard cell phones and deaf users operating the application.

Technical Deep-Dive: Multi-Stream Data and Signaling Split

When a standard user calls the network, the incoming audio packet stream is captured by a cloud gateway server and converted into a raw linear pulse-code modulation (PCM) stream. The audio is routed to a high-speed automatic speech recognition engine (Whisper AI) to isolate text phrases, which then activate corresponding 3D sign animations on the deaf user's screen. Conversely, the deaf user's outbound camera stream is handled entirely on the local device, avoiding costly server-side video uploads. Only the matched text tokens are transmitted to the cloud, where they instantly generate neural Swahili audio payloads that are fed back into the caller's active phone connection.

5. DETAILED SYSTEM ARCHITECTURE LAYER SPECIFICATIONS

Architecture Layer	Functional Components & Engineering Operations	Protocols & Libraries Used
1. Media Capture Layer	Handles local mobile device camera capture, audio loop buffers, interface view rendering, and microphone permissions.	Flutter Camera Plugin / WebRTC MediaDevices API
2. Outbound Vision Engine	Processes spatial-temporal camera matrices, executes few-shot classification using CLIP, and outputs Swahili tokens.	ONNX Mobile Engine / PyTorch Lite
3. Voice Routing Layer	Checks the global user profile setting state (Male/Female) and appends speech metadata parameters onto outbound texts.	Dart State Provider / BLoC Framework
4. Inbound Audio Parser	Captures incoming call audio payloads, processes voice inputs via automatic speech recognition (ASR), and produces text strings.	OpenAI Whisper API / Google Cloud ASR
5. Graphic Animation Layer	Takes inbound text streams, references language dictionaries, and renders clean 3D skeletal animations.	Unity Embedded Widget / WebGL Vector Matrix
6. Telephony Bridge	Manages connection handshakes, encapsulates network media packets, and bridges mobile data channels with phone carrier networks.	SIP Signaling Protocols / Twilio Voice Voice SDK

6. ARCHITECTURAL ADVANTAGES & STRATEGIC IMPACT

- 1. Two-Way Communication Autonomy:** The system eliminates the need for third-party human interpreters, enabling total conversational independence during both everyday face-to-face meetings and active telephone calls.
- 2. Bandwidth & Data Efficiency:** Local processing of the deaf user's video feed avoids the high data consumption of server-side uploads. Only lightweight text tokens are transmitted over the network, ensuring stable operation even over slower 3G connections.
- 3. Acoustic Self-Expression:** By leveraging manual voice profile overrides, the application respects the user's gender identity and personal presentation preference, surmounting the structural gender neutralities inherent to Swahili grammar.

KEY SUPPORTING SCIENTIFIC REFERENCES

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